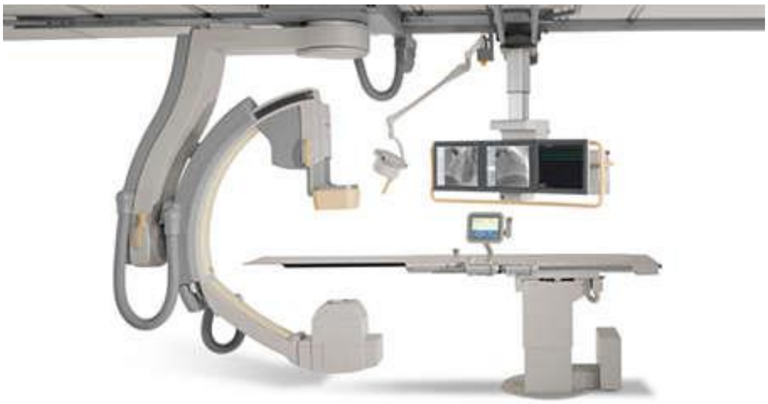
Model-driven quality and resource Management for CPSs

Freek van den Berg f.g.b.v.d.berg@tue.nl



The Fitoptivis Project

A Cyber-Physical System (CPS) integrates cyber systems, human users, networks and physical systems. Therefore, a CPS needs visual context and awareness to make autonomous and correct decisions.

However, advanced image and video processing is computationally intensive and challenging. Also, a CPS comprises increasingly complex and distributed configurations, which is reflected in the growing number of sensors, actuators and other smart devices. To make matters worse, a CPS needs to simultaneously satisfy many rigorous constraints, e.g., real-time behaviour, safety, quality, and performance

In the Fitoptivis project, we aim to develop a reference architecture that can be used to derive many feasible designs of a CPS. After this, a selection of the optimal designs is made.

The Philips Healthcare case study

We investigate aspects of the interventional X-Ray (iXR) systems of our project partner Philips.

An iXR system (see Figure above) assists a surgeon during an operation, viz., it provides a continuous stream of images of the inside of a patient's body on a display. These images are based on X-ray beams and require deep image processing to be interpretable by the surgeon. In the case study, we focus on the situation in which multiple streams of images need to be shown in the display. This inherently leads to situations in which parallel processing and prioritizing play a role.

Bibliography

Freek van den Berg was born in Nijmegen, The Netherlands, in 1980. He received his BSC and MSc, both cum laude, in Information Science at Radboud University Nijmegen. In 2012, he started a PhD titled ``Performance evaluation of component-based systems" at the University of Twente in Enschede in collaboration with Philips Healthcare. This resulted into a PhD in 2017. After this, he pursued one year of PostDoc at the University of Twente in collaboration with Wageningen University and Research with a focus on CPS. At the beginning of 2019, he attained a PostDoc position titled ``model-driven quality and resource management for CPSs" in the aforementioned Fitoptivis project.

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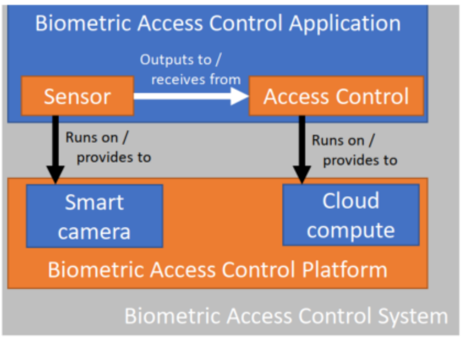

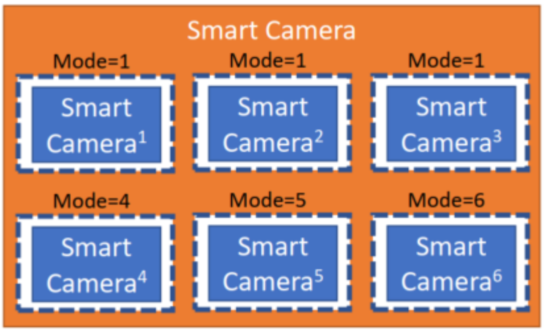

The Conceptual Model QRML

We present the conceptual model of the component-based Quality and Resource Management Language (QRLM). Such components come in three flavors:

Atomic component which has input and output (I/O) dependencies, and provided-budget and required budget dependencies;

Composite component which consists of a number of components, dependencies among them, and external dependencies;

Configurable component which can behave in different ways using different configurations. A configuration is represented as a mapping from parameters to a component that exhibits the configured behavior.

At its highest level of abstraction, the model consists of one component which can be refined arbitrarily often using the composite and configurable component constructs. At the lowest level of abstraction, there are only atomic components.

An illustrative QRML instance

To illustrate how QRML is put into practice, we model a Biometric Access Control System (BAC) in QRML. A **BAC system** (see left figure) holds the door closed for intruders and allows authorized people, which is determined via biometric authentication, access to resources.

A BAC system comprises the following four atomic components: (i) **Sensor** detects events in its environment; (ii) **Access Control** limits access to resources; (iii) **Smart Camera** extracts application-specific information from images; and, (iv) **Cloud Compute** is an on-demand computer system resource. Furthermore, **BAC application** is composition of Sensor and Access Control and **BAC platform** of Smart Camera and Cloud Compute. The figure shows that the Sensor application component runs on the Smart camera platform and Access Control runs on Cloud Compute. They are related by budget dependencies. Also, Sensor outputs to Access Control, an I/O dependency.

Next, we refine component **Smart Camera** to have six configurations (see right figure). The configurations map a mode to components which different frame-rates and resolutions.

Automated design support in QRML

Automated design support analyses the BAC system for each configuration choice of the Smart Camera. Since Sensor runs on Smart Camera, the choice is constrained by the budget requirement of Sensor.